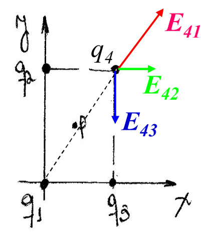
1)



$q_1 = 2q \ \langle 0,0\rangle \qquad q_3 = -q \ \langle a,0\rangle$

$q_2 = q \ \langle 0,b\rangle \qquad q_4 = -3q \ \langle a,b\rangle$

A) $\vec{E}_4 = \vec{E}_{41} + \vec{E}_{42} + \vec{E}_{43}$

$\vec{E}_{41} = \frac{k\,q_1\,\hat{r}_{41}}{r_{41}^2}$; $\vec{r}_{41} = \vec{r}_4 - \vec{r}_1 = \langle a,b\rangle - \langle 0,0\rangle = \langle a,b\rangle$ m

$r_{41} = |\vec{r}_{41}| = \sqrt{a^2+b^2}$ m

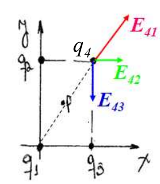
$\hat{r}_{41} = \frac{\vec{r}_{41}}{|r_{41}|} = \frac{\langle a,b\rangle}{\sqrt{a^2+b^2}}$

$\vec{E}_{41} = \frac{k\,2q}{(a^2+b^2)}\,\frac{\langle a,b\rangle}{\sqrt{a^2+b^2}} = 2kq\,\frac{\langle a,b\rangle}{(a^2+b^2)^{3/2}}$ [N/C]

1

$\vec{E}_{42} = k\,\frac{q_2\,\hat{r}_{42}}{r_{42}^2}$; $\vec{r}_{42} = \langle a,b\rangle - \langle 0,b\rangle = \langle a,0\rangle$ [m]

$r_{42} = a$; $\hat{r}_{42} = \langle 1,0\rangle$



$\vec{E}_{42} = \frac{k\,q}{a^2}\cdot\langle 1,0\rangle$ [N/C]

$\vec{E}_{43} = \frac{k\,q_3\,\hat{r}_{43}}{r_{43}^2} = \frac{k\,(-q)\,\vec{r}_{43}}{r_{43}^3}$; $\vec{r}_{43} = \langle a,b\rangle - \langle a,0\rangle = \langle 0,b\rangle$ [m]

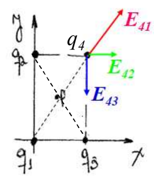
$r_{43} = b$ [m] ; $\hat{r}_{43} = \langle 0,1\rangle$

$\vec{E}_{43} = \frac{k\,(-q)}{b^2}\cdot\langle 0,1\rangle$ [N/C]

$\vec{E}_4 = \frac{2kq\,\langle a,b\rangle}{(a^2+b^2)^{3/2}} + \frac{kq}{a^2}\langle 1,0\rangle - \frac{kq}{b^2}\langle 0,1\rangle$ [N/C]

2

$$\vec{E}_4 = \left\{\left[\frac{2kqa}{(a^2+b^2)^{3/2}} + \frac{kq}{a^2}\right]\hat{\imath} + \left[\frac{2kqb}{(a^2+b^2)^{3/2}} - \frac{kq}{b^2}\right]\hat{\jmath}\right\}\left[\frac{N}{C}\right]$$



La fuerza sobre la carga q_4 es:

$$\vec{F}_4 = \vec{F}_{41} + \vec{F}_{42} + \vec{F}_{43} = q_4\vec{E}_4$$

$$\vec{F}_4 = (-3q)\vec{E}_4 = \left\{-\left[\frac{6kq^2a}{(a^2+b^2)^{3/2}} + \frac{3kq^2}{a^2}\right]\hat{\imath} - \left[\frac{6kq^2b}{(a^2+b^2)^{3/2}} - \frac{3kq^2}{b^2}\right]\hat{\jmath}\right\}[N]$$

B) ¿$W_{\infty P}$? Por definición: $V_P = \frac{W_{\infty P}}{q_5}$;

$$\left(\text{Nota}: V_P - \cancelto{0}{V_\infty} = \frac{W_{\infty P}}{q_5}\right)$$

$$V_P = V_{P1} + V_{P2} + V_{P3} + V_{P4}$$

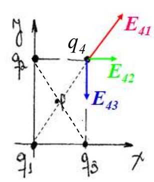
$P\langle \frac{a}{2} ; \frac{b}{2} \rangle$

$V_{P1} = \frac{kq_1}{r_{P1}}$; $\vec{r}_P - \vec{r}_1 = \vec{r}_{P1} = \langle \frac{a}{2} ; \frac{b}{2} \rangle - \langle 0,0 \rangle = \langle \frac{a}{2}, \frac{b}{2} \rangle$

$$r_{P1} = \sqrt{\frac{a^2}{4} + \frac{b^2}{4}} = \frac{\sqrt{a^2+b^2}}{2}$$

3

$$V_{P1} = \frac{k \cdot 2q}{\sqrt{a^2+b^2}} \cdot 2 = \frac{4kq}{\sqrt{a^2+b^2}} \; [V]$$



Por simetría: $r_{P1} = r_{P2} = r_{P3} = r_{P4}$

$$\therefore \; V_{P2} = \frac{kq_2}{r_{P2}} = \frac{kq \cdot 2}{\sqrt{a^2+b^2}} = \frac{2kq}{\sqrt{a^2+b^2}} \; [V]$$

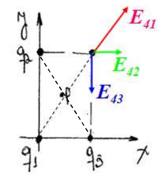
$$V_{P3} = \frac{kq_3}{r_{P3}} = \frac{k(-q)}{\frac{\sqrt{a^2+b^2}}{2}} = -\frac{2kq}{\sqrt{a^2+b^2}} \; [V]$$

$$V_{P4} = \frac{kq_4}{r_{P4}} = \frac{k(-3q)}{\frac{\sqrt{a^2+b^2}}{2}} = -\frac{6kq}{\sqrt{a^2+b^2}}$$

$$V_P = \frac{4kq}{\sqrt{a^2+b^2}} + \frac{2kq}{\sqrt{a^2+b^2}} - \frac{2kq}{\sqrt{a^2+b^2}} - \frac{6kq}{\sqrt{a^2+b^2}} = -\frac{2kq}{\sqrt{a^2+b^2}} \; [V]$$

4

$$W_{\infty P} = -\frac{2kq}{\sqrt{a^2+b^2}} \cdot 5q = -\frac{10kq^2}{\sqrt{a^2+b^2}} \ [J]$$

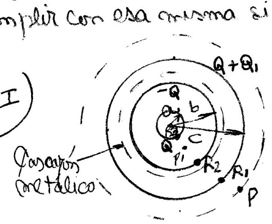


2) A) $\oint \vec{E} \cdot d\vec{A} = \frac{q_{neta}}{\varepsilon_0}$

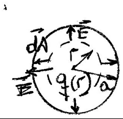
i) Para calcular el Φ_E sólo se necesita utilizar una superficie cerrada imaginaria. $\Phi_E = \frac{q_{neta}}{\varepsilon_0}$

ii) Para poder hallar el $\vec{E}$ usando la ley de Gauss, la distribución de las cargas debe poseer un alto grado de simetría. La superficie utilizada deberá cumplir con esa misma simetría.

B) I)

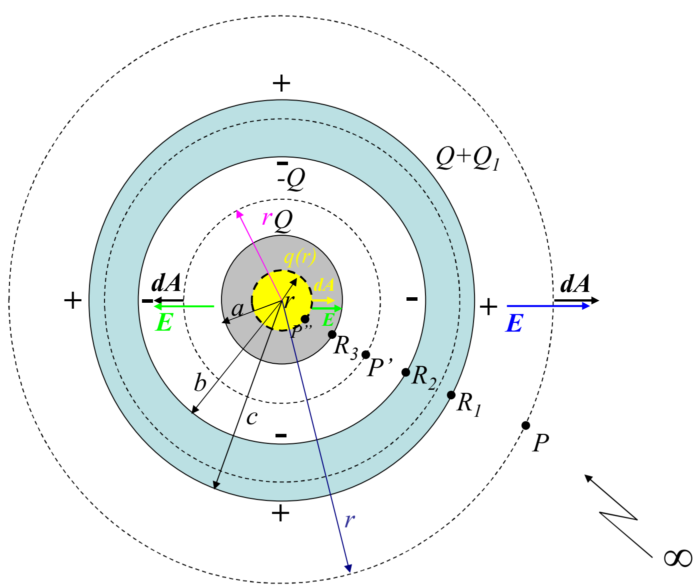


Para $0 \le r < a$



Debemos hallar $q(r)$

5



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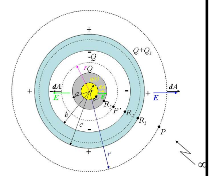
Para la superficie gaussiana de radio r la carga encerrada es $q(r)$

$$q(r) = \rho \cdot \frac{4}{3}\pi r^3 \quad ; \quad \oint \vec{E} \cdot \vec{dA} = \oint E . dA . \cos 0 = E \oint dA = E . 4\pi r^2$$

$$E . \cancel{4\pi} r^{\cancel{2}} = \frac{q(r)}{\varepsilon_0} = \frac{\rho}{\varepsilon_0} \cdot \frac{4}{3} \cancel{\pi} r^3 \Rightarrow E(r) = \frac{\rho . r}{3\varepsilon_0} \; [N/C]$$



Para $a < r < b$

La carga encerrada es $q(r=a) = \rho \cdot \frac{4}{3}\pi a^3 = Q$

$$E(r) = \frac{kQ}{r^2} = \frac{1}{\cancel{4\pi}\varepsilon_0} \frac{\rho \cdot \frac{\cancel{4}}{3}\pi a^3}{r^2} = \frac{\rho a^3}{3\varepsilon_0 r^2} \; [N/C]$$

Para $b < r < c$ $\vec{E} = 0$ por el principio fundamental de la electrostática.

Para $r > c$ la carga encerrada es $(Q + Q_1)$

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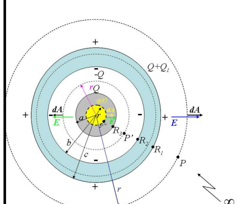
$$E(r) = \frac{k(Q+Q_1)}{r^2} = \frac{1}{4\pi\varepsilon_0} \frac{(Q+Q_1)}{r^2} \; [N/C]$$

Resumiendo :

$$E_r(r) = \begin{cases} \frac{\rho r}{3\varepsilon_0} & 0 \leq r < a \\ \frac{\rho a^3}{3\varepsilon_0 r^2} & a < r < b \\ 0 & b < r < c \\ \frac{1}{4\pi\varepsilon_0} \frac{(Q+Q_1)}{r^2} & r > c \end{cases}$$



II) Adoptamos como referencia de potencial el punto impropio ∞. Allí tomamos $V_\infty = 0 \, V$. Para $r \geq c$:

$$V_P - \cancel{V_\infty}^{0} = -\int_\infty^r E_r \, dr = -\int_\infty^r \frac{1}{4\pi\varepsilon_0} \frac{(Q+Q_1)}{r^2} = \frac{1}{4\pi\varepsilon_0} \frac{(Q+Q_1)}{r} \; [V]$$

$$V(r) = V_P = \frac{k(Q+Q_1)}{r} \; [V]$$

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$$V_{R1} = V(r=c) = \frac{k(Q+Q_1)}{c} \; [V]$$

Para $b \leq r < c$ el potencial es constante e igual a V_{R1}

$$V_{R2} = V_{R1}.$$

Para $a \leq r < b$ $\quad V_{P'} - V_{R2} = -\int_b^r \frac{\rho . a^3}{\varepsilon_0 r^2 3} dr = \frac{\rho a^3}{3\varepsilon_0}\left(\frac{1}{r} - \frac{1}{b}\right) [V]$

$$V_{R3} = \frac{\rho . a^3}{3\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right) + V_{R1}$$

Para $0 \leq r < a$ $\quad V_{P''} - V_{R3} = -\int_a^r \frac{\rho r}{3\varepsilon_0} dr = -\frac{\rho . (r^2 - a^2)}{6\varepsilon_0}$

$$V_{P''} = \frac{\rho}{6\varepsilon_0}\left(a^2 - r^2\right) + \frac{\rho}{3\varepsilon_0}\left(a^2 - \frac{a^3}{b}\right) + V_{R1}$$

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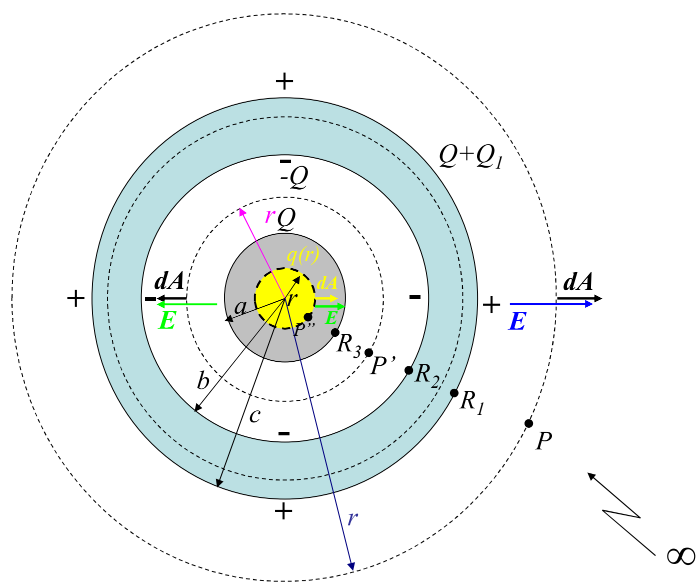

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$$V(r) = \begin{cases} \frac{\rho}{6\varepsilon_0}(a^2 - r^2) + \frac{\rho}{3\varepsilon_0}\left(a^2 - \frac{a^3}{b}\right) + \frac{k(Q+Q_1)}{c} & 0 \leq r < a \\ \frac{\rho a^3}{3\varepsilon_0}\left(\frac{1}{r} - \frac{1}{b}\right) + \frac{k(Q+Q_1)}{c} & a \leq r < b \\ \frac{k(Q+Q_1)}{c} & b \leq r < c \\ \frac{k(Q+Q_1)}{r} & r \geqslant c \end{cases}$$

III) Como para $r = 2c$ el $V(2c) = 360\ V$.

$$\frac{k.(Q+Q_1)}{2c} = 360\ V$$

$$Q = \rho \cdot \frac{4}{3}\pi a^3 = 62.10^{-6} \cdot \frac{4}{3}\pi \cdot (0{,}03)^3 = 7.10^{-9}\,[C]$$

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Por tanto, $Q_1 = \frac{360 \cdot 2c}{9 \cdot 10^9} - Q = \frac{360 \cdot 2 \cdot 0{,}1}{9 \cdot 10^9} - 7.10^{-9} = 1.10^{-9}\,[C]$

Con el valor de Q y de Q_1 podríamos hallar tanto $E(r)$ como $V(r)$ en forma más compacta.

$$E(r) = \begin{cases} 2{,}33.10^{6}.r & 0 \leq r < a \\ \frac{63{,}05}{r^2} & a \leq r < b \\ 0 & b < r < c \\ \frac{72}{r^2} & r > c \end{cases}$$

$$\underset{[V]}{V(r)} = \begin{cases} 1{,}167.10^{6}(a^2 - r^2) + 2033{,}56 & (\text{si } 0 \leq r < a) \\ 63{,}05\left(\frac{1}{r} - \frac{1}{b}\right) + 720 & (\text{si } a \leq r < b) \\ 720 & \text{si } b \leq r < c \\ \frac{72}{r} & \text{si } r > c \end{cases}$$

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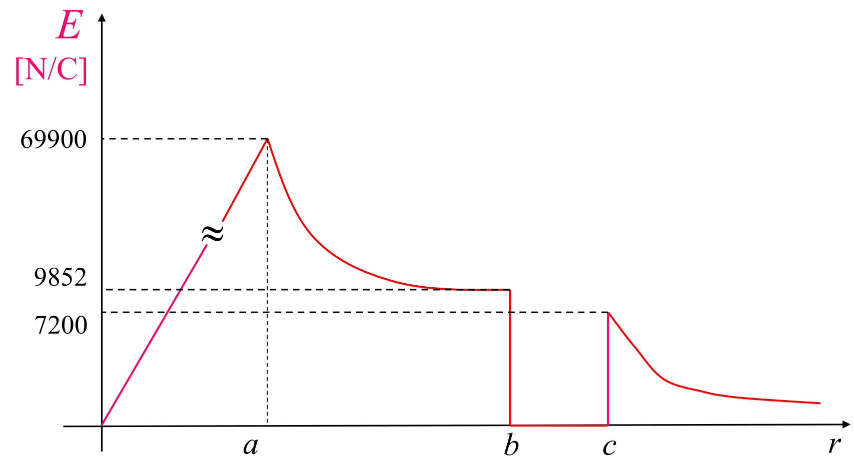
Gráfica de E vs r



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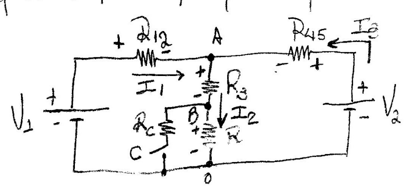
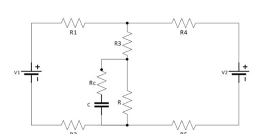
3) Cuando el capacitor se cargó completamente, actúa como una llave abierta. El circuito equivalente sería :



$R_{12} = R_1 + R_2 = 16\Omega$

$R_{45} = R_4 + R_5 = 24\Omega$



Aplicamos las reglas de Kirchoff.

$$\left.\begin{array}{l} V_1 - I_1 \cdot R_{12} - (R_3 + R) I_2 = 0 \\ V_2 - I_3 R_{45} - (R_3 + R) I_2 = 0 \\ I_1 + I_3 = I_2 \end{array}\right\} \Rightarrow \begin{array}{l} 16 I_1 + 12 I_2 = 20 \\ 12 I_2 + 24 I_3 = 12 \\ I_1 - I_2 + I_3 = 0 \end{array}$$

Resolviendo el sistema de ecuaciones :

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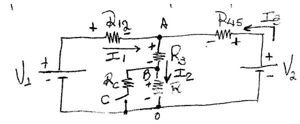
$I_1 = 0{,}667\ A$

$I_2 = 0{,}778\ A$

$I_3 = 0{,}111\ A$



La potencia disipada en las resistencias es:

$$P_{R_1} = I_1^2 \cdot R_1 = (0{,}667)^2 \cdot 10 = 4{,}45\ W$$

$$P_{R_2} = I_1^2 \cdot R_2 = (0{,}667)^2 \cdot 6 = 2{,}67\ W$$

$$P_{R_3} = I_2^2 \cdot R_3 = (0{,}778)^2 \cdot 8 = 4{,}84\ W$$

$$P_{R} = I_2^2 \cdot R = (0{,}778)^2 \cdot 4 = 2{,}42\ W$$

$$P_{R_4} = I_3^2 \cdot R_4 = (0{,}111)^2 \cdot 14 = 0{,}17\ W$$

$$P_{R_5} = I_3^2 \cdot R_5 = (0{,}111)^2 \cdot 10 = 0{,}12\ W$$

$$P_{\text{Total, disipada}} = 14{,}67\ W$$

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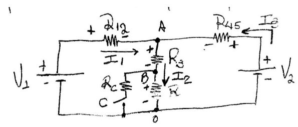
La potencia entregada por las baterías es:



$$P_{V_1} = V_1 \cdot I_1 = 20 \cdot 0{,}667 = 13{,}34\ W$$

$$P_{V_2} = V_2 \cdot I_3 = 12 \cdot 0{,}111 = 1{,}33\ W$$

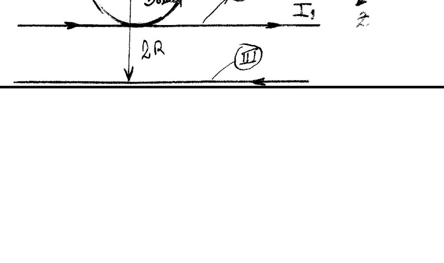
$$P_{\text{total entregada}} = 14{,}67\ W$$

$U_{cap} = \frac{1}{2} C \cdot V_{cap}^2$; $\quad V_{cap} = V_{B0} = V_B - V_0 = I_2 \cdot R$

$V_{cap} = 0{,}778 \cdot 4 = 3{,}11\ V$

$$U_{cap} = \frac{1}{2} \cdot 81 \cdot 10^{-6} \cdot (3{,}11)^2 = 3{,}91 \cdot 10^{-4}\ J$$

4)



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Tenemos en el origen de coordenadas tres campos magnéticos, los que son generados por los tres tramos de conductores.

Tramo I) Para la espira circular, el campo generado en el punto 0 es:

(Biot-Savart) $\vec{B}_{0I} = \frac{\mu_0 I_1}{2R}(\hat{k}) = \frac{4\pi.10^{-7}.7}{2.0{,}02}.\hat{k} = 2{,}2.10^{-4}\hat{k}\,[T]$

Tramo II) Para el conductor muy largo que transporta I_1, el campo generado se halla aplicando la ley de Ampere.

$$\vec{B}_{0II} = \frac{\mu_0 I}{2\pi R}\hat{k} = \frac{\cancel{4}^{2}\cancel{\pi}.10^{-7}.7}{\cancel{2}\cancel{\pi}.\cancel{0{,}02}_{0{,}01}}\hat{k} = 7.10^{-5}\hat{k}\,[T]$$

Tramo III) Idem al anterior $\vec{B}_{0III} = \frac{\mu_0 I_2}{2\pi(2R)}(-\hat{k})$

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$$\vec{B}_{0III} = \frac{\cancel{4}\cancel{\pi}.10^{-7}.I_2}{\cancel{2}\cancel{\pi}.\cancel{2}.0{,}02}(-\hat{k}) = 5.10^{-6}I_2(-\hat{k})$$

$\therefore$ $\vec{B}_0 = \vec{B}_{0I} + \vec{B}_{0II} + \vec{B}_{0III}$

$\vec{B}_0 = 2{,}2.10^{-4}\hat{k} + 7.10^{-5}\hat{k} - 5.10^{-6}I_2.(-\hat{k}) = 0$

$\therefore$ $I_2 = 58\text{ A}$.

5) B) La masa de las partículas α es $m_\alpha = 4m_p$

En la zona de aceleración se cumple que:

$\Delta E_k + \Delta E_p = 0$; $\frac{1}{2}m_\alpha v_f^2 = q_\alpha.\Delta V \Rightarrow v_f = \left(\frac{q_\alpha.\Delta V.2}{m_\alpha}\right)^{1/2}$

$$v_f = \left(\frac{2.q_\alpha\,\Delta V}{m_\alpha}\right)^{1/2} = \left(\frac{\cancel{2}.\cancel{2}.q_p.\Delta V}{\cancel{4}m_p}\right)^{1/2} = \left(\frac{1{,}6.10^{-19}.25000}{1{,}67.10^{-27}}\right)^{1/2}$$

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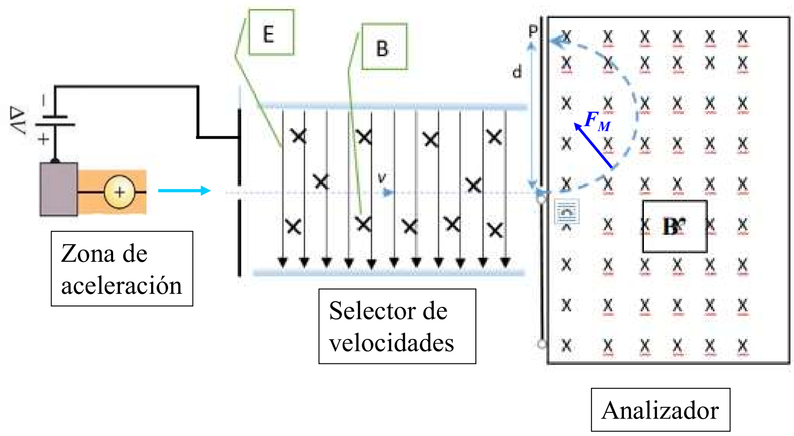
ESPECTRÓMETRO DE MASAS



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$v_f = 1{,}548 \cdot 10^6 \frac{m}{s}$

c) Dado que $\vec{F}_M + \vec{F}_E = 0$ $\quad \cancel{q_\alpha} v_f B = \cancel{q_\alpha} E$

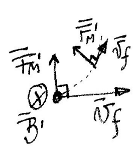
$$v_f \cdot B = E \Rightarrow B = \frac{E}{v_f} = \frac{2500}{1{,}548 \cdot 10^6} = 1{,}61 \cdot 10^{-3} \ [T]$$

D) Dado que $F_M' = F_{cent}$

$$\cancel{q_\alpha} \cancel{v_f} \cdot B' = \frac{m_\alpha v_f^{\cancel{2}}}{R} \rightarrow B' = \frac{m_\alpha \cdot v_f}{q_\alpha \cdot R} = \frac{4 \cdot m_p \cdot v_f}{\cancel{2} q_p \cdot \frac{d}{\cancel{2}}}$$

$$B' = \frac{4 \cdot 1{,}67 \cdot 10^{-27} \cdot 1{,}548 \cdot 10^{6}}{1{,}6 \cdot 10^{-19} \cdot 0{,}5} = 0{,}129 \ T$$

$$\vec{F}_M' = q_\alpha \cdot (\vec{v}_f \times \vec{B}')$$



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